

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA16 **COURSE CODE:** Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO2: Apply suitable Data Pre processing techniques like Data cleaning, Data integration, Data transformation and data Reduction ,for effective data preparation (BL3,BL4)
Assessment Tool Weightage: 50%

QUESTIONS: 5

Total Marks:50

Annexure B
Analytical Problem Solving

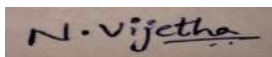
<i>Criteria</i>	Problem Understanding (2)	Method Selection (2)	Solution Process (2.5)	Accuracy of Calculation (2)	Interpretation of Results (1.5)	<i>Total(10)</i>
<i>Weightage</i>	20%	20%	25%	20%	15%	100%
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						
<i>Q4</i>						
<i>Q5</i>						

Code of Conduct:

I _____ N.Vijetha (192311258) _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION:

Numerical Problems					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Manhattan distance (L₁ form): $d(i, j) = |x_i - x_j|$

(ii) Euclidean distance: $\sqrt{(x_i - x_j)^2} = d(i, j)$

(iii) Supremum

— ACTIVITY 04

① Suppose that the data for analysis includes the attributes age. The age values for the data types are (in increasing order) 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70

(a) what is the mean of the data? what is the median?

(b) what is the mode of the data?

comment on the data's modality (i.e., bimodal, trimodal, etc)

(c) what is the midrange of the data?

(d) can you find (roughly) the first quartile (Q_1) and the third quartile (Q_3) of the data?

(e) Give the five-number summary of the given data?

(f) show a boxplot of the data.

(g) Give quantile-by-quantile plot graph

$$(a) \text{ Mean} = \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

$$= \frac{13+15+16+16+19+20+20+21+22+22+25+25+25+25+30+33+33+35+35+35+35+36+40+45+46+52+70}{27}$$

$$= \frac{809}{27}$$

$$= 29.9629$$

$$\approx 30$$

$$= \frac{n+1}{2} = \frac{27+1}{2} = \frac{28}{2}$$

$$= 14^{\text{th}} \text{ value}$$

$$\text{Median} = 25$$

(b) mode = 25 & 35 are coming continuously
 - As only 2 values are repeating, they are called as bimodal.

(c) Mid range =
$$\frac{\text{max value} + \text{min value}}{2}$$

$$= \frac{70 + 13}{2} = \frac{83}{2} = 41.5$$

(d) - First quartile:

$Q_1 = 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25$

- median of $Q_1 = \frac{13+1}{2} = \frac{14}{2} = 7^{\text{th}} \text{ value}$

median $\Rightarrow 20$

- Third quartile:

$Q_3 = 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70$

- median of $Q_3 = \frac{33+1}{2} = \frac{14}{2} = 7^{\text{th}} \text{ value}$

median $\Rightarrow 35$

$\therefore$ first quartile (Q_1) = 20

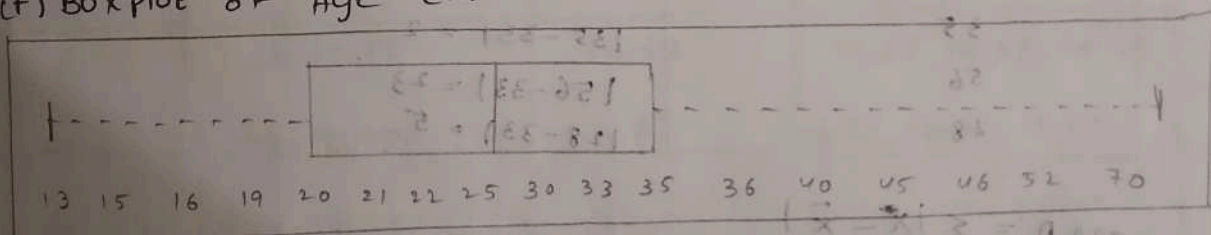
Third quartile (Q_3) = 35

(e) Five number Summary:

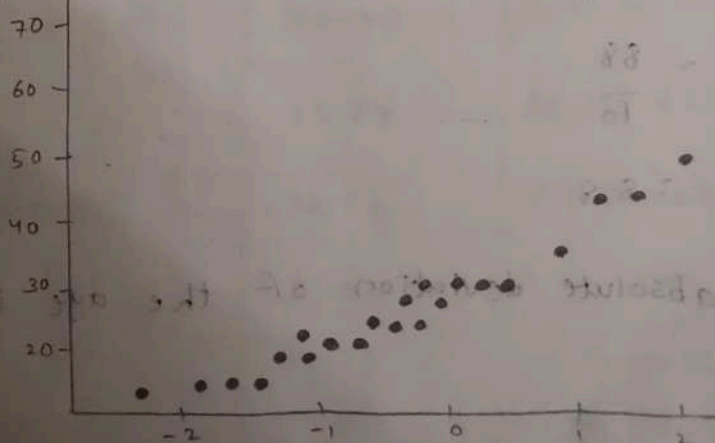
$\rightarrow$ min, Q_1 , median, Q_3 , max

$\rightarrow 13, 20, 25, 35, 70$

(f) Boxplot of Age data:



(g) Q-Q plot graph:



② Given the following measurements for the variable by the following: standardize the age.

Compute the mean absolute deviation of the age.

- Age: 18, 22, 25, 42, 28, 43, 33, 35, 56, 28

- mean:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

$$= \frac{18 + 22 + 25 + 42 + 28 + 43 + 33 + 35 + 56 + 28}{10}$$

$$= \frac{330}{10} = 33$$

- mean absolute deviation of the age:

- mean = 33

$x(\text{age})$	$ x - \bar{x} $
18	$ 18 - 33 = 15$
22	$ 22 - 33 = 11$
25	$ 25 - 33 = 8$
42	$ 42 - 33 = 9$
28	$ 28 - 33 = 5$
43	$ 43 - 33 = 10$
33	$ 33 - 33 = 0$
35	$ 35 - 33 = 2$
56	$ 56 - 33 = 23$
28	$ 28 - 33 = 5$

$$MAD = \frac{\sum |x - \bar{x}|}{n}$$

$$= \frac{88}{10}$$

$$= 8.8$$

∴ mean absolute deviation of the age is 8.8

③ Suppose that the data for analysis includes the attribute age. The age values for the data tuples are (in increasing order) 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 30, 33, 33, 35, 35, 35, 36, 40, 45, 46, 52, 70. Use smoothing by bin means to smooth the above data, using a bin depth of 3. Illustrate your steps.

- No. of values = 27

- Bin depth = 3

$$\text{count of Bin} = \frac{\text{occurrence}}{\text{Bin depth}} = \frac{27}{3} = 9$$

$$\text{Bin 1} = 13, 15, 16 = \text{mean} = \frac{13+15+16}{3} = 14.6666$$

$$\text{Bin 2} = 16, 19, 20 = \text{mean} = \frac{16+19+20}{3} = 18.3333$$

$$\text{Bin 3} = 20, 21, 22 = \text{mean} = \frac{20+21+22}{3} = 21$$

$$\text{Bin 4} = 22, 25, 25 = \text{mean} = \frac{22+25+25}{3} = 24$$

$$\text{Bin 5} = 25, 25, 30 = \text{mean} = \frac{25+25+30}{3} = 26.6666$$

$$\text{Bin 6} = 33, 33, 35 = \text{mean} = \frac{33+33+35}{3} = 33.6666$$

$$\text{Bin 7} = 35, 35, 35 = \text{mean} = \frac{35+35+35}{3} = 35$$

$$\text{Bin 8} = 36, 40, 45 = \text{mean} = \frac{36+40+45}{3} = 40.3333$$

$$\text{Bin 9} = 46, 52, 70 = \text{mean} = \frac{46+52+70}{3} = 56$$

- Smooth Bin: Replace the mean for the all values in the Bin.

Bin	original value	Bin mean	Smoothed values
B ₁	13, 15, 16	14.67	14.67, 14.67, 14.67
B ₂	16, 19, 20	18.33	18.33, 18.33, 18.33
B ₃	20, 21, 22	21.00	21.00, 21.00, 21.00
B ₄	22, 25, 25	24.00	24.00, 24.00, 24.00
B ₅	25, 25, 30	26.67	26.67, 26.67, 26.67
B ₆	33, 33, 35	33.67	33.67, 33.67, 33.67
B ₇	35, 35, 35	35.00	35.00, 35.00, 35.00
B ₈	36, 40, 45	40.33	40.33, 40.33, 40.33
B ₉	46, 52, 70	56.00	56.00, 56.00, 56.00

④ Use the two methods below to normalize the following group of data: 200, 300, 400, 600, 1000

① min-max normalization by setting $\min=0$ & $\max=1$

② z-score normalization

(a) min-max Normalization ($\min=0$, $\max=1$)

$$x' = \frac{x - \min(x)}{\max(x) - \min(x)}$$

$$x' = \frac{x - 200}{1000 - 200} = \frac{x - 200}{800}$$

Original value (x)	calculation	Normalized value
200	$\frac{200-200}{800} = 0/800$	0.00
300	$\frac{300-200}{800} = 100/800$	0.125
400	$\frac{400-200}{800} = 200/800$	0.25
600	$\frac{600-200}{800} = 400/800$	0.50
1000	$\frac{1000-200}{800} = 800/800$	1.00

$\therefore$ min-max Normalization = (0.00, 0.125, 0.25, 0.50, 1.00)

(b) z-score Normalization

$$z = \frac{x - \mu}{\sigma}$$

(i) mean, $\mu = \frac{200 + 300 + 400 + 600 + 1000}{5} = \frac{2500}{5} = 500$

(ii) Standard deviation:

x	$x - \mu$	$(x - \mu)^2$
200	$200 - 500 = -300$	$(-300)^2 = 90000$
300	$300 - 500 = -200$	$(-200)^2 = 40000$
400	$400 - 500 = -100$	$(-100)^2 = 10000$
600	$600 - 500 = 100$	$(100)^2 = 10000$
1000	$1000 - 500 = 500$	$(500)^2 = 250000$

$$\sum (x - \mu)^2 = 90000 + 40000 + 10000 + 10000 + 250000$$

$$= 400000$$

$$\text{variance} = \sigma^2 = \frac{400000}{5} = 80000$$

$$\text{standard deviation} = \sigma = \sqrt{\frac{\sum (x - \mu)^2}{n}}$$

following

$$\sigma = \sqrt{80000}$$

$$\sigma \approx 282.84$$

(1) compute z-scores:

original value	calculate	z-score
200	$(200-500)/282.84$	-1.06
300	$(300-500)/282.84$	-0.71
400	$(400-500)/282.84$	-0.35
600	$(600-500)/282.84$	0.35
1000	$(1000-500)/282.84$	1.77

z-scores are -1.06, -0.71, -0.35, 0.35, 1.77

Suppose a group of 12 sales price records has been sorted as follows: 5, 10, 11, 13, 15, 35, 50, 55, 72, 92, 204, 215

Partition them into three bins by each of the following methods

(a) Equal-Frequency Partitioning

(b) Equal-width partitioning

(c) clustering

- Total Records = 12

- Total Bins = 3

(a) Equal-frequency partitioning

- No. of Records per bin = $\frac{12}{3}$
= 4 values

- Bin 1 = 5, 10, 11, 13

- Bin 2 = 15, 35, 50, 55

- Bin 3 = 72, 92, 204, 215

(b) Equal-width partitioning

- width = $\frac{\text{maximum} - \text{minimum}}{\text{number of bins}}$

$$= \frac{215 - 5}{3}$$

$$= \frac{210}{3}$$

The intervals are: first value + 70 - 1

• Bin 1 : 5-74

• Bin 2 : 75-144

• Bin 3 : 145-215

Now place the values into the intervals

Bin	Interval	values
Bin 1	5-74	5, 10, 11, 13, 15, 35, 50, 55, 72
Bin 2	75-144	92
Bin 3	145-215	204, 215

(c) clustering:

↳ clustering groups similar values together while separating distant values

A natural clustering of data is:

cluster	values
cluster 1	5, 10, 11, 13, 15
cluster 2	35, 50, 55, 72, 92
cluster 3	204, 215

(6) suppose a hospital tested the age & body fat data for 18 randomly selected adults with the following result.

age	23	23	27	27	39	41	47	49	50
% fat	9.5	26.5	7.8	17.8	31.4	25.9	27.4	27.2	31.2

age	52	54	54	56	57	58	58	60	61
% fat	34.6	42.5	28.8	33.4	30.2	34.1	32.9	41.2	35.7

(a) calculate the mean, median & standard deviation of age & % fat

(b) Draw the box plots for age & % fat

(c) Draw a scatter plot & a q-q plot based on these 2 variables

(d) normalize the 2 variables based on z-score Normalization.

(e) calculate the correlation coefficient (person's product moment coefficient). Are these 2 variables positively (or) negatively correlated?

(a) mean, median & standard deviation:

measure	age	% fat
mean	46.44	28.78
median	51	30.70
standard deviation	13.22	9.25

Age:

- mean:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

$$= \frac{23 + 23 + 27 + 27 + 39 + 41 + 47 + 49 + 50 + 52 + 54 + 54 + 56 + 57 + 58 + 58 + 60 + 61}{18}$$

$$= \frac{836}{18} = 46.44$$

- median: $n = 18$ (Even)

→ Average of 9th & 10th values

→ 9th value = 50 → 10th value = 52

$$\Rightarrow \frac{50 + 52}{2} = 51$$

- standard deviation:

$$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$$

$$\text{mean} = 46.44$$

$$\rightarrow \text{SD} = 13.22$$

- % fat:

$$\text{- mean: } \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

$$= \frac{9.5 + 26.5 + 7.8 + 17.8 + 31.4 + 25.9 + 27.4 + 27.2 + 31.2 + 34.6 + 42.5 + 28.8 + 33.4 + 30.2 + 34.1 + 32.9 + 41.2 + 35.7}{18}$$

$$= \frac{518.1}{18}$$

$$= 28.78$$

- Median: $n=18$ (even)
- Average of 9th & 10th values

$$= \frac{30.2 + 31.2}{2} = 30.7$$

- Standard deviation:

$$\text{Mean} = 28.78$$

$$SD = \sqrt{\frac{\sum (x - \bar{x})^2}{n-1}}$$

$$= 9.25$$

⑤ Box-plot for age

- Five number summary: (age)

$$\text{Minimum} = 23$$

$$\text{Maximum} = 61$$

$$\text{Median } (Q_2) \rightarrow \frac{18}{2} = 9$$

$$9^{\text{th}} \text{ value} = 50$$

$$10^{\text{th}} \text{ value} = 52$$

$$\text{Median} = \frac{50 + 52}{2}$$

$$= \frac{102}{2} = 51$$

- First Quartile (Q_1)

$$23, 23, 27, 27, 39, 41, 47, 49, 50$$

$$\text{Median} = \frac{9+1}{2} = 5$$

$$5^{\text{th}} \text{ value} = 39$$

$$\therefore Q_1 = 39$$

- Third quartile (Q_3)

$$52, 54, 54, 56, 57, 58, 58, 60, 61$$

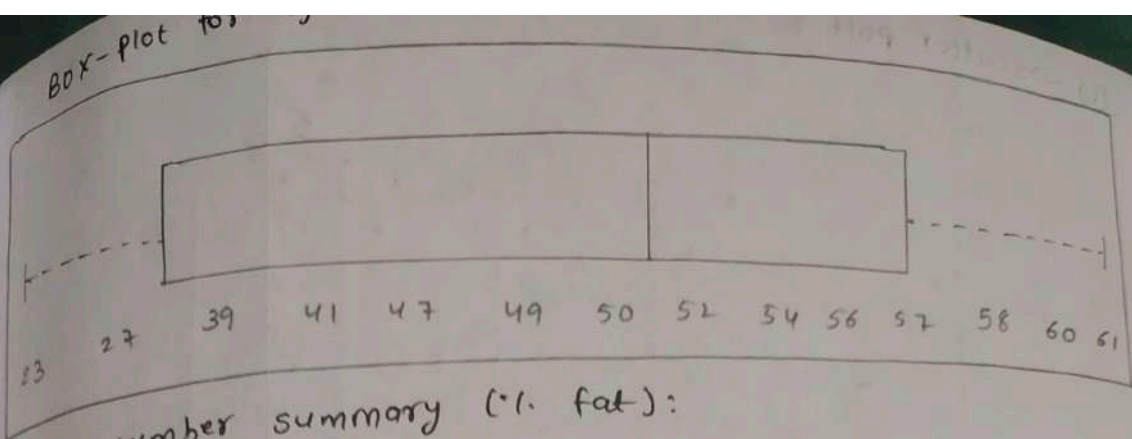
$$\text{Median} = \frac{9+1}{2} = 5$$

$$5^{\text{th}} \text{ value}$$

$$= 57$$

$$\therefore Q_3 = 57$$

$\therefore$ Five number summary:
Min, max, Q_1 , Q_3 , Median
23, 61, 39, 57, 51



- Five number summary (% fat):

- minimum = 7.8, - maximum = 42.5

- median $\rightarrow$ 18 observations, 9th value = 30.2
10th value = 31.2

$$\rightarrow \frac{30.2 + 31.2}{2} = \frac{61.4}{2} = 30.7$$

- First quartile (Q_1):

7.8, 9.5, 17.8, 25.9, 26.5, 27.2, 27.4, 28.8, 30.2

- 9 observations

$$-\frac{n+1}{2} = \frac{10}{2} = 5^{\text{th}} \therefore Q_1 = 26.5$$

- Third quartile (Q_3):

31.2, 31.4, 32.9, 33.4, 34.1, 34.6, 35.7, 41.2, 42.5

- 9 observations

$$-\frac{n+1}{2} = \frac{10}{2} = 5^{\text{th}}$$

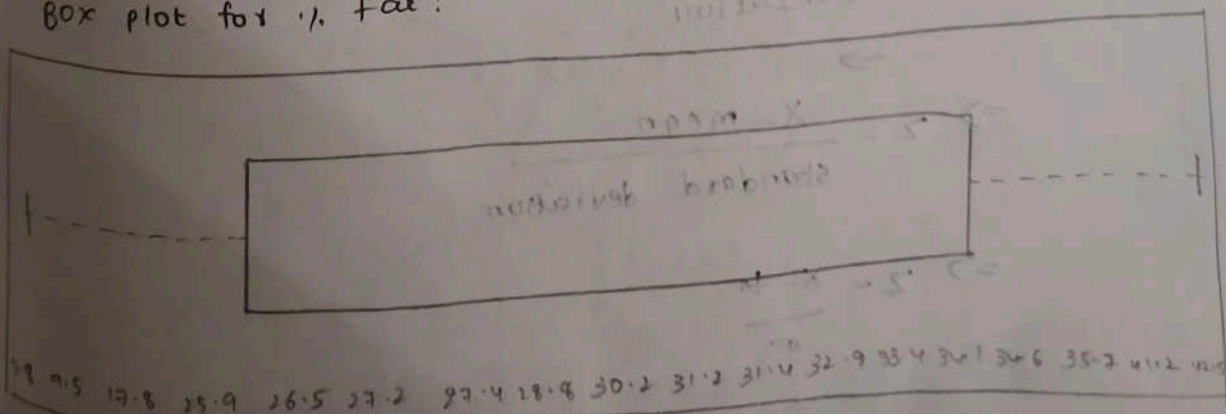
$$\therefore Q_3 = 34.1$$

- Five number summary:

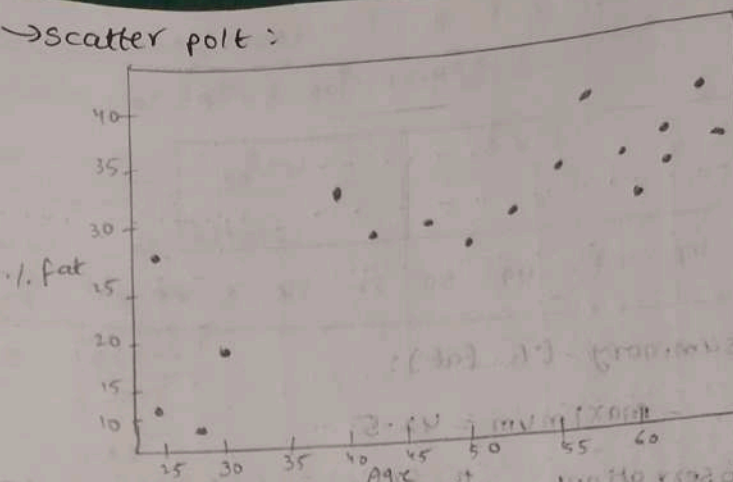
- min, Q_1 , median, Q_3 , max

- 7.8, 26.5, 30.7, 34.1, 42.5

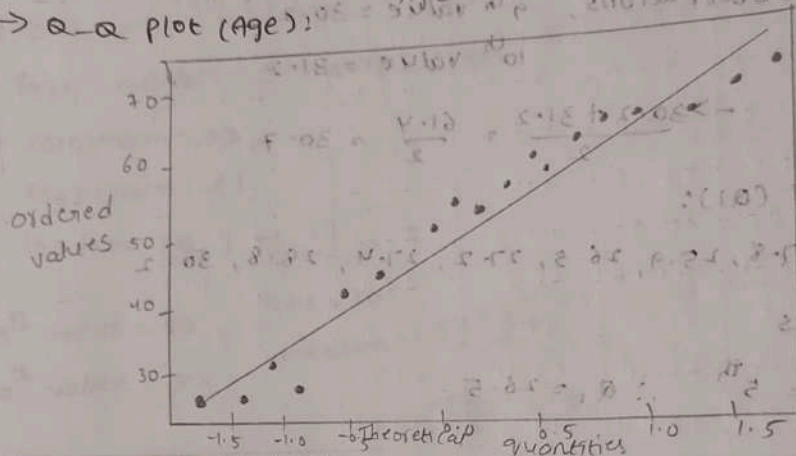
Box plot for % fat:



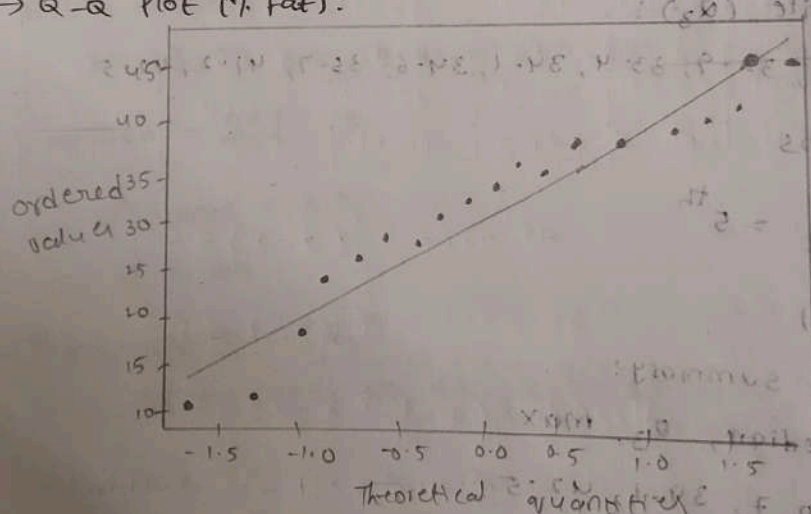
(c) → scatter plot:



→ Q-Q plot (Age):



→ Q-Q plot (% fat):



(d) Z-score normalization:

$$\Rightarrow Z = \frac{X - \text{mean}}{\text{standard deviation}}$$

$$\Rightarrow Z = \frac{X - M}{\sigma}$$

⇒ Z-score

Age	Z-score
23	23
23	27
27	27
27	3
39	
41	
47	
49	
50	
52	
54	
54	
56	
57	
58	
58	
60	
61	

⇒ Z-score

% fat	Z-score
9.5	
26.5	
7.8	
17.8	
31.4	
25.9	
27.4	
27.2	
31.2	
34.6	
42.5	
28.8	
38.4	
30.2	
34.1	
32.9	
41.2	
25.7	

Age	calculation	z-score
23	$23 - 46.44 / 13.22 = -23.44 / 13.22$	-1.77
23	$23 - 46.44 / 13.22 = -23.44 / 13.22$	-1.77
27	$27 - 46.44 / 13.22 = -19.44 / 13.22$	-1.47
27	$27 - 46.44 / 13.22 = -19.44 / 13.22$	-1.47
39	$39 - 46.44 / 13.22 = -7.44 / 13.22$	-0.56
39	$41 - 46.44 / 13.22 = -5.44 / 13.22$	-0.41
41	$47 - 46.44 / 13.22 = 0.56 / 13.22$	0.04
47	$49 - 46.44 / 13.22 = 2.56 / 13.22$	0.19
49	$50 - 46.44 / 13.22 = 3.56 / 13.22$	0.27
50	$52 - 46.44 / 13.22 = 5.56 / 13.22$	0.42
52	$54 - 46.44 / 13.22 = 7.56 / 13.22$	0.57
54	$54 - 46.44 / 13.22 = 7.56 / 13.22$	0.57
54	$56 - 46.44 / 13.22 = 9.56 / 13.22$	0.72
56	$57 - 46.44 / 13.22 = 10.56 / 13.22$	0.80
57	$58 - 46.44 / 13.22 = 11.56 / 13.22$	0.87
58	$58 - 46.44 / 13.22 = 11.56 / 13.22$	0.87
58	$60 - 46.44 / 13.22 = 13.56 / 13.22$	1.03
60	$61 - 46.44 / 13.22 = 14.56 / 13.22$	1.10
61		

% fat	calculation	z-score
9.5	$9.5 - 28.78 / 9.25 = -19.28 / 9.25$	-2.08
16.5	$26.5 - 28.78 / 9.25 = -2.28 / 9.25$	-0.25
27.8	$7.8 - 28.78 / 9.25 = -20.98 / 9.25$	-2.27
17.8	$17.8 - 28.78 / 9.25 = -10.98 / 9.25$	-1.19
31.4	$31.4 - 28.78 / 9.25 = 2.62 / 9.25$	0.28
25.9	$25.9 - 28.78 / 9.25 = -2.88 / 9.25$	-0.31
27.4	$27.4 - 28.78 / 9.25 = -1.38 / 9.25$	-0.15
27.2	$27.2 - 28.78 / 9.25 = -1.58 / 9.25$	-0.17
31.2	$31.2 - 28.78 / 9.25 = 2.42 / 9.25$	0.26
34.6	$34.6 - 28.78 / 9.25 = 5.82 / 9.25$	0.63
42.5	$42.5 - 28.78 / 9.25 = 13.72 / 9.25$	1.48
28.8	$28.8 - 28.78 / 9.25 = 0.02 / 9.25$	0.00
32.4	$32.4 - 28.78 / 9.25 = 3.62 / 9.25$	0.39
30.2	$30.2 - 28.78 / 9.25 = 1.42 / 9.25$	0.15
34.1	$34.1 - 28.78 / 9.25 = 5.32 / 9.25$	0.58
32.9	$32.9 - 28.78 / 9.25 = 4.12 / 9.25$	0.44
41.2	$41.2 - 28.78 / 9.25 = 12.42 / 9.25$	1.34
35.7	$35.7 - 28.78 / 9.25 = 6.92 / 9.25$	0.75

e) Pearson's product moment correlation coefficient:

$$r = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2 \sum (y - \bar{y})^2}}$$

age ← % fat mean →

Age (x)	% Fat (y)	$x - \bar{x}$	$y - \bar{y}$	$(x - \bar{x})^2$	$(y - \bar{y})^2$	$(x - \bar{x})(y - \bar{y})$
23	9.5	-23.44	-19.28	549.43	371.72	451.54
23	26.5	-23.44	-2.28	549.43	5.20	53.44
27	7.8	-19.44	-20.98	377.91	440.16	407.85
27	17.8	-19.44	-10.98	377.91	120.56	213.45
39	31.4	-7.44	2.62	55.35	6.86	-19.49
41	25.9	-5.44	-2.88	29.59	8.29	15.67
47	27.4	0.56	-1.38	0.31	1.90	-0.77
49	27.2	2.56	-1.58	6.55	2.50	-4.04
50	31.2	3.56	2.42	12.68	5.86	8.62
52	34.6	5.56	5.82	30.91	33.87	32.36
54	42.5	7.56	13.72	57.15	188.24	103.7
54	28.8	7.56	0.02	57.15	0.00	0.15
56	33.4	9.56	4.62	91.39	21.34	44.17
57	30.2	10.56	1.42	111.51	2.02	15.00
58	34.1	11.56	5.32	133.63	28.30	61.50
58	32.9	11.56	4.12	133.63	16.97	47.63
60	41.2	13.56	12.42	183.87	154.26	168.42
61	35.7	14.56	6.92	211.99	47.89	100.76

$$\Rightarrow \sum (x - \bar{x})^2 = 3360.94$$

$$\Rightarrow \sum (y - \bar{y})^2 = 1453.36$$

$$\Rightarrow \sum (x - \bar{x})(y - \bar{y}) = 1805.16$$

- substitute in the formula!

$$r = \frac{1805.16}{\sqrt{3360.94 \times 1453.36}}$$

$$= \frac{1805.16}{\sqrt{4884035.64}}$$

$$= \frac{1805.16}{2209.98}$$

$$= 0.817 \approx$$

Pearson

correlation coefficient

$$r = 0.818$$